



IIC Majuro Geoid:  
Development of a Geoid for Majuro  
Atoll, Republic of Marshall Islands

Assignment Title:

**LiDAR Survey Data Quality Assurance  
and Quality Control**

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## Abstract

IIC Technologies (IIC), under contract to The Pacific Community (SPC), was contracted to provide Quality Control and Quality Assurance of contractor provided topographic and bathymetric LiDAR Data of the Republic of the Marshall Islands (RMI). As part of this QC process, IIC investigated the source of an approximately 1m vertical offset between control from four surveys between 2014 and 2017. The resulting report (*IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020*) showed the primary cause of the discrepancy was the 2016 survey's adopted coordinate for the continuously operating GNSS reference station MAJU. It also identified that the geoid developed for Majuro was based on this 2016 survey and that additional inconsistencies existed.

As a result, SPC commissioned IIC to undertake this Review in order to further assess the documentation and data from the 2014 to 2017 surveys in order to establish a revised geoid for Majuro. The results of the Review are contained within this Report.

This review has concluded that the 2014, 2015 and 2017 surveys were, for the most part, consistent with each other. However, even after the 1m offset at MAJU was accounted for, inconsistencies in the 2016 control point ellipsoidal heights remained (20cm to 40cm). Further, the inconsistencies in the primary control were propagated into the subsequent 2016 RTK survey.

The 2014, 2015, and 2017 ellipsoid and orthometric heights were found to be consistent with a vertically adjusted EGM08 geoid. The ellipsoid heights from these surveys were also consistent with the 2019 LiDAR data.

Rather than attempt to adjust the existing geoid, a new geoid was created by vertically adjusting the EGM08 geoid. This new geoid matched the observed geoid (ellipsoid minus ortho heights at control points) with a mean offset of 3cm and a standard deviation of 5cm (not including the 2016 survey control).

The final geoid provides a model to translate between ellipsoid heights, referenced to ITRF2014 epoch 2016, and orthometric heights, referenced to MSL as defined by the 2020 Tidal prediction manual as 1.102 m above tide gauge zero.

This report provides the findings of the review and information on the creation and validation of the new geoid "Mgeoid\_ITRF14ep2016\_MGVD17".

This Report builds on the findings and recommendations in the *IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020*. Therefore, it is recommended that both documents be read in conjunction.



This analysis and the resulting Report were undertaken and prepared by Dr David Dodd.

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## 1 Introduction

IIC Technologies (IIC), under contract to The Pacific Community (SPC), was contracted to provide Quality Control and Quality Assurance of contractor provided topographic and bathymetric LiDAR Data of the Republic of the Marshall Islands (RMI). As part of that review, and at SPC request, IIC investigated the source of an approximately 1m vertical offset between control from four surveys between 2014 and 2017. The resulting report (*IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020*) showed the primary cause of the discrepancy was the 2016 survey's adopted coordinate for the continuously operating GNSS reference station MAJU. It also identified that the geoid developed for Majuro was based on this 2016 survey and that additional inconsistencies existed.

Therefore, SPC requested IIC to further review both the documentation and data from the 2014 to 2017 surveys in order to identify the issues and establish a revised geoid for Majuro; the findings of which are contained in this Report.

This review has concluded that the 2014, 2015 and 2017 surveys were, for the most part, consistent with each other. However, even after the 1m offset at MAJU was accounted for, inconsistencies in the 2016 control point ellipsoidal heights remained (20cm to 40cm). These inconsistencies in the primary control propagated into the subsequent 2016 RTK survey. Therefore, a new geoid has been developed from the Earth Gravity Model of 2008, created and distributed by the US National Geospatial-Intelligence Agency (NGA), see: [https://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/egm08\\_wgs84.html](https://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/egm08_wgs84.html). The model was vertically shifted to match the geoid height at the tide gauge holding BM MAR13.

**This report provides the findings of the review and information on the creation and validation of the new geoid "Mgeoid\_ITRF14ep2016\_MGVD17".**

*It is noted that this Report does not supersede but builds on the findings and recommendations in the IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020. Therefore, both documents should be read together.*

## 2 Orthometric Datum

The orthometric relationship between many Bench Marks (BMs) on the East side of Majuro was established in a 2009 trigonometric (total station) levelling campaign. The results of the levelling survey were within the required 1mm/km accuracy level. BMs are shown in Figure 1 and relative elevations are shown in Figure 2 (Taken from <https://www.sonel.org/spip.php?page=nivellement&idStation=2117>).

The orthometric height of the GNSS station MAJU and its BM (MAJUBM) were included in the levelling campaign, as were MAR3, MAR13, MAR31 and MAR107, all of which were important to the establishment of the vertical control for the tide gauge. These BMs were also used to establish the orthometric heights of the other BMs on Majuro.

A temporary tide gauge was established on Majuro for approximately 7 months in 1968 and 1969. MAR3 was the holding BM for the 1968/1969 temporary tide gauge and was the holding elevation for the 2009 levelling work. Also, the 2020 tidal prediction document references tidal measurements to MAR3.

The geodetic datum relationship between tide gauge zero (TGZ) of the new gauge, RLR (Revised Local Reference) and the BMs is shown in Figure 2.

For the purpose of this report, orthometric heights referenced to the 1968/69 MSL datum are referred to as Majuro Geodetic Vertical Datum 1969 (MGVD69).

RLR to TGZ = 5.941m, TGZ to MGVD69 = 1.051m

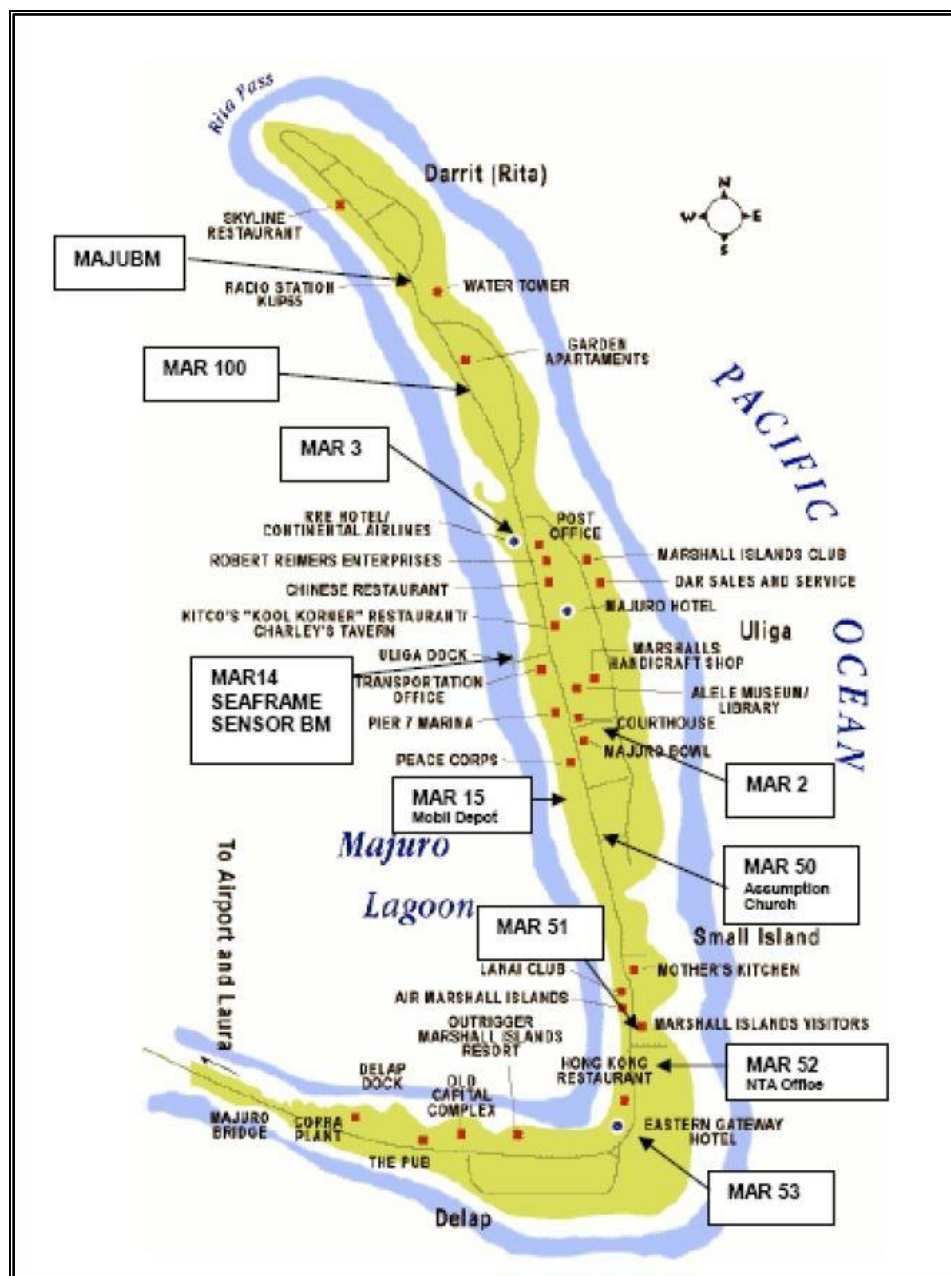


Figure 1: Majuro BMs. Re 2009 leveling campaign

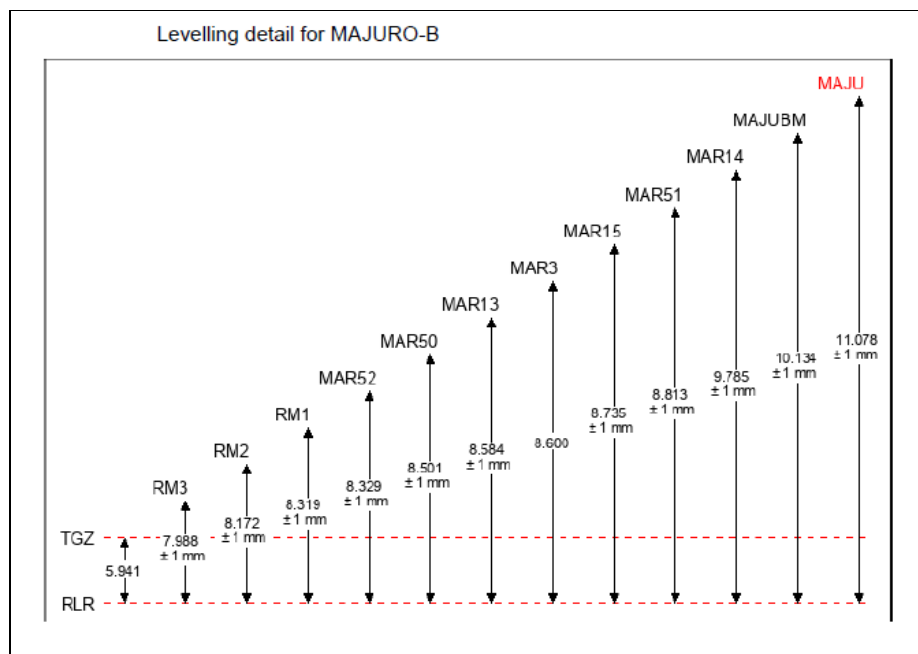


Figure 2: Levelling detail for MAJURO-B TG (Sonel station 2117)., from 2009 levelling campaign.  
Taken from <https://www.sonel.org/spip.php?page=nivellement&idStation=2117>

Unfortunately, BM MAR3 is now inside a building and cannot have GNSS observed directly on it. Also, it is approximately 500 m north of the existing TG, as such it is not a viable holding BM for the new gauge.

The primary holding BM for the new gauge is MAR13 and it has GNSS observations from the 2017 NOAA survey. The other BMs in the vicinity are MAR31 (steel screw in concrete) and MAR107 (deep Driven BM). MAR14 is on the TG pipe and does not have an ellipsoid height.

Ellipsoid heights for several of the BMs were established during the 2014 to 2017 GNSS surveys. BMs MAR13, MAR31 and MAR107 had ellipsoid heights established during the 2017 survey. This survey also established a new vertical datum for the BMs in relation to TG Zero (see Figure 3). This datum was established at 1.088m above TGZ, 37mm higher than MGVD69.

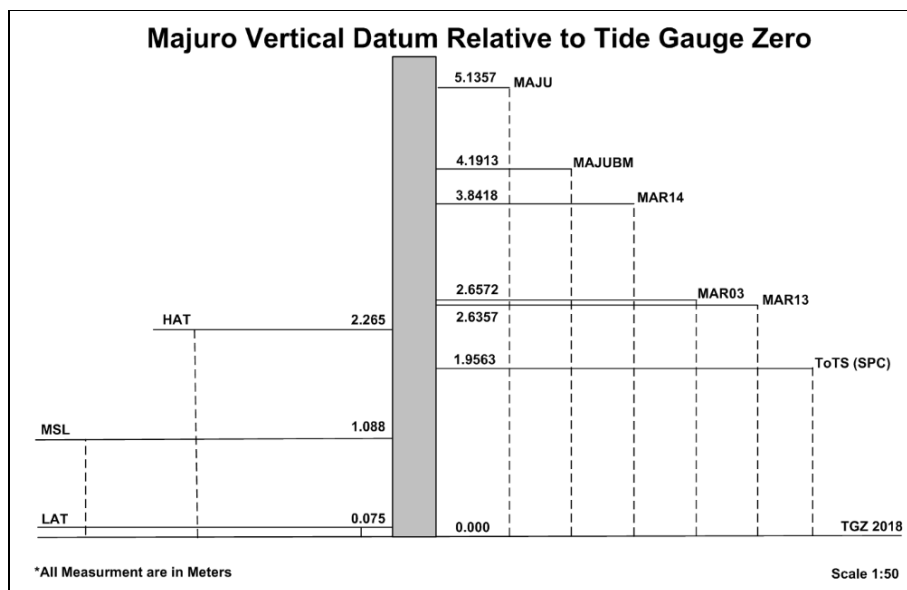


Figure 3: Wiring diagram from 2017 NOAA survey

Table 1 below shows several primary BMs referenced in both the 2009 survey and the 2017 NOAA survey. TGZ 2009 compares well to TGZ of 2017, all at 2mm or less, except MAR13 which differs by 7 mm. LMSL from 2017 will be used in this report to compare to 2016 orthometric heights.

**Note: The NOAA 2017 MSL datum is discussed here to establish the orthometric elevations with respect to MGVD69. This datum is not used beyond this point.**

|               | RLR<br>(SONEL) | TGZ<br>(SONEL) | TGZ<br>(2017) | LMSL<br>(2017) | MGVD69       |
|---------------|----------------|----------------|---------------|----------------|--------------|
| <b>MAJU</b>   | 11.078         | 5.137          | 5.136         | 4.048          | <b>4.086</b> |
| <b>MAJUBM</b> | 10.134         | 4.193          | 4.191         | 3.103          | 3.142        |
| <b>MAR14</b>  | 9.785          | 3.844          | 3.842         | 2.754          | 2.793        |
| <b>MAR3</b>   | 8.600          | 2.659          | 2.657         | 1.569          | 1.608        |
| <b>MAR13</b>  | 8.584          | 2.643          | 2.636         | 1.548          | 1.592        |
| <b>MAR31</b>  | 8.792          | 2.851          | 2.846         | <b>1.758</b>   | <b>1.800</b> |
| <b>MAR107</b> | 8.657          | 2.716          | 2.712         | 1.624          | 1.665        |

Table 1: Elevation summary of main BMS, 2009 and 2017 comparison in metres

The geodetic datum used here for the development of the new geoid is the MSL value established for the 2020 tidal prediction document (see <http://www.bom.gov.au/ntc/IDO60054/IDO60054.2020.pdf>). The MSL datum was established as the mean tide from 1993 to 2017 = 1.102m above TGZ (see Figure 4). For this discussion, this datum is referred to as MGVD17. MGVD17 is 0.051m higher than MGVD69

The 2020 tidal prediction document references MAR3. It is suggested that future publications refer to one of the holding BMs, such as MAR13 (see Figure 5).

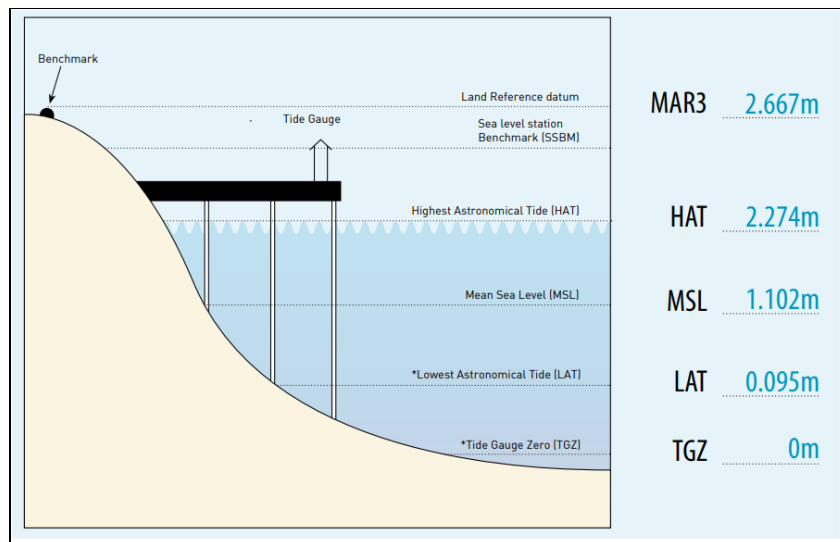


Figure 4: MGVD17

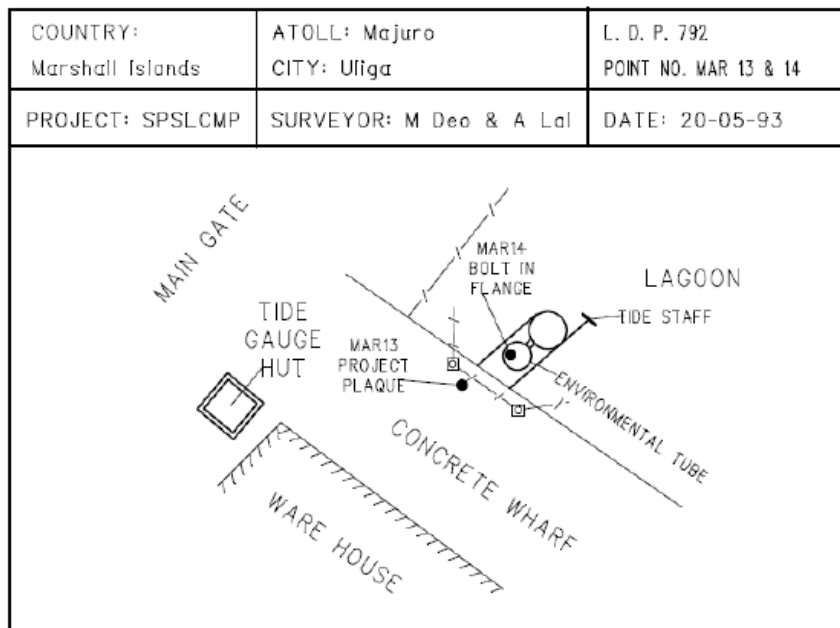


Figure 5: MAR13 and MAR14 site description

In summary, elevations of BMs relative to MVVD69 and MGVD17 are as tabulated below:

|               | MGVD69 | MGVD17 |
|---------------|--------|--------|
| <b>MAJU</b>   | 4.086m | 4.035m |
| <b>MAJUBM</b> | 3.142m | 3.091m |
| <b>MAR14</b>  | 2.793m | 2.742m |
| <b>MAR3</b>   | 1.608m | 1.557m |
| <b>MAR13</b>  | 1.592m | 1.541m |
| <b>MAR31</b>  | 1.800m | 1.749m |
| <b>MAR107</b> | 1.665m | 1.614m |

*Table 2: BM Elevations*

### 3 Common Points

One of the biggest challenges to the verification of the existing geoid was determining what information to accept, and what to view as suspect. To do this, IIC reviewed control points common to multiple surveys. IIC also relied heavily on an ellipsoidally referenced version of the 2019 LiDAR Geotiff. The LiDAR data provided ellipsoid heights of the terrain at control point locations and only provided valid comparison if the control points were also at ground level. Therefore, the LiDAR information was used as corroborating evidence as opposed to concrete proof. The following three points were evaluated in detail; Peace Park 2002 in the South, MAR31 in the East, and Lauraes 2002 in the West.

#### **Peace Park 2002 (South)**

The control point “**Peace Park 2002**” was referenced in both the 2014 and 2015 surveys. Both used exactly the same coordinates. The 2016 survey used the same location for control point “BS16/maj\_base\_092316”.

The coordinates put Peace Park and BS16/Maj\_Base\_092316 approximately 0.3 m apart, horizontally. From the 2016 survey Panel point PNL73 is approximately 13m from the Peace PaBS16/Maj\_Base\_0923166 control point (see Figure 6).

Table 3 shows the index number, station name, latitude, longitude, orthometric height, ellipsoid height and the LiDAR derived ellipsoidal height interpolated from the 1m Geotiff image, and the observed geoid value N (N = ellipsoidal height minus orthometric height).

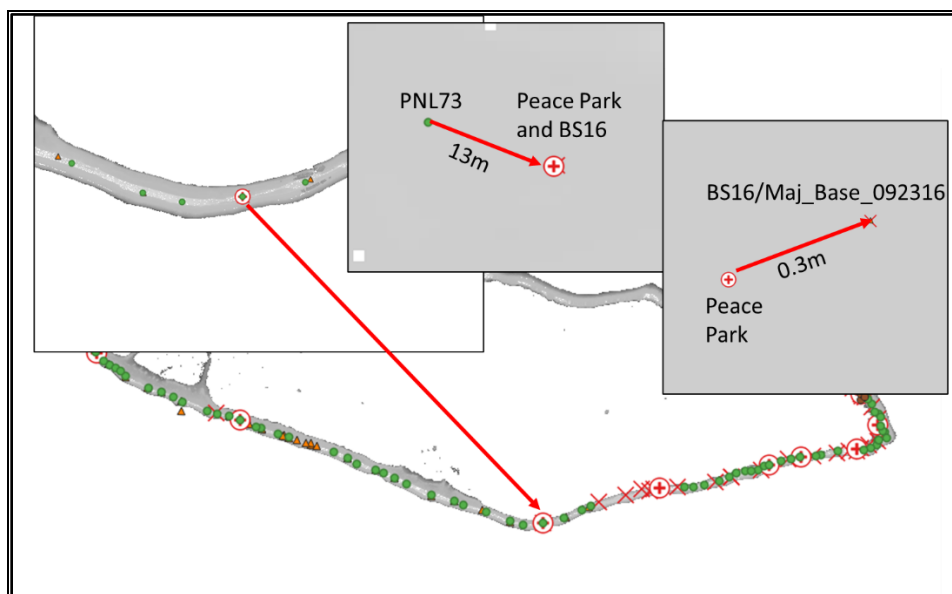


Figure 6: Peace Park 2002, BS16/Maj\_Base\_092316 and PNL73 location

| ID         | Survey | STN                  | Orth<br>(MGVD69) | Ell Ht<br>(AusPos) | LiDAR<br>Ell Ht | N (Ell-<br>Ortho) |
|------------|--------|----------------------|------------------|--------------------|-----------------|-------------------|
| 201400031  | 2014   | Peace Park 2002      | 1.605m           | 31.516m            | 31.464m         | 29.911m           |
| 201500057  | 2015   | Peace Park 2002      | 1.605m           | 31.516m            | 31.464m         | 29.911m           |
| 2016000201 | 2016   | BS16/maj_base_092316 | 1.605m           | 31.316m            | 31.464m         | 29.711m           |
| 2016000144 | 2016   | PNL73                | 1.46m            | 31.173m            | 31.389m         | 29.723m           |

Table 3: Peace Park 2002, BS16/Maj\_Base\_092316 and PNL73 height comparison

The Lidar derived heights for Peace Park 2002 and BS16/Maj\_Base\_092316 were the same because they were at the same location. The orthometric height was also the same for all three control points. However, the ellipsoid height for BS16 was 20cm lower. As a result, the observed geoid value differed by 20cm. The 2019 LiDAR ellipsoid height was 5cm lower than the 2014 and 2015 heights, and 15cm higher than the 2016 BS16.

From the 2016 survey PNL73 was likely derived from BS16. The 2019 LiDAR ellipsoid height was 22cm higher than PNL73, similar to the control point discrepancy.

### MAR31 (East)

The BM MAR31, located on the East side of Majuro (see

Figure 7), was included in the 2014, 2016 and 2017 surveys. The orthometric heights from 2014 and 2016 were identical, and the 2017 survey height was 6mm lower (see

Table 4). The ellipsoidal heights were all different, but within 34mm of each other. The 2019 LiDAR ellipsoid height was within approximately 10cm of the control ellipsoidal heights.

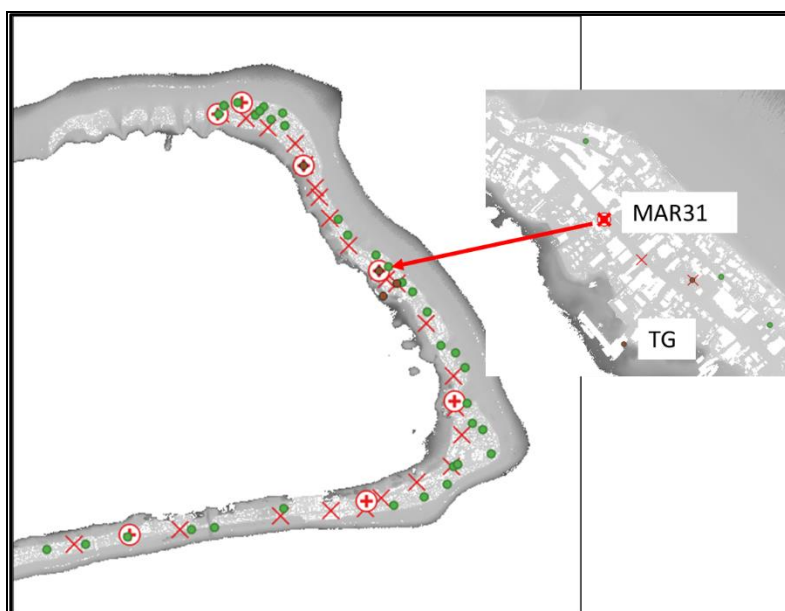


Figure 7: MAR31

| ID         | Survey | STN                  | Ortho Ht (MGVD69) | Ell Ht (AusPos) | LiDAR Ell Ht | N (Ell – Ortho) |
|------------|--------|----------------------|-------------------|-----------------|--------------|-----------------|
| 201400011  | 2014   | MAR31                | 1.801m            | 31.443m         | 31.337m      | 29.642m         |
| 2016000188 | 2016   | BS01/maj_base_091416 | 1.801m            | 31.409m         | 31.337m      | 29.608m         |
| 2016000204 | 2017   | MAR31                | 1.795m            | 31.434m         | 31.337m      | 29.639m         |

Table 4: MAR31 height comparison

### Lauraes 2002 (West)

Lauraes 2002 was used in the 2014 and 2015 surveys and their heights (orthometric and ellipsoidal) were identical (see Figure 8 and

Table 5). The ellipsoidal heights were 13cm higher than the 2019 LiDAR ellipsoidal heights.

PNL319 (2016 survey) was located approximately 20m away. Its ellipsoid height was 28cm lower than the LiDAR data.

The observed geoid value (N) was 27 cm lower at PNL319 than at Lauraes 2002.

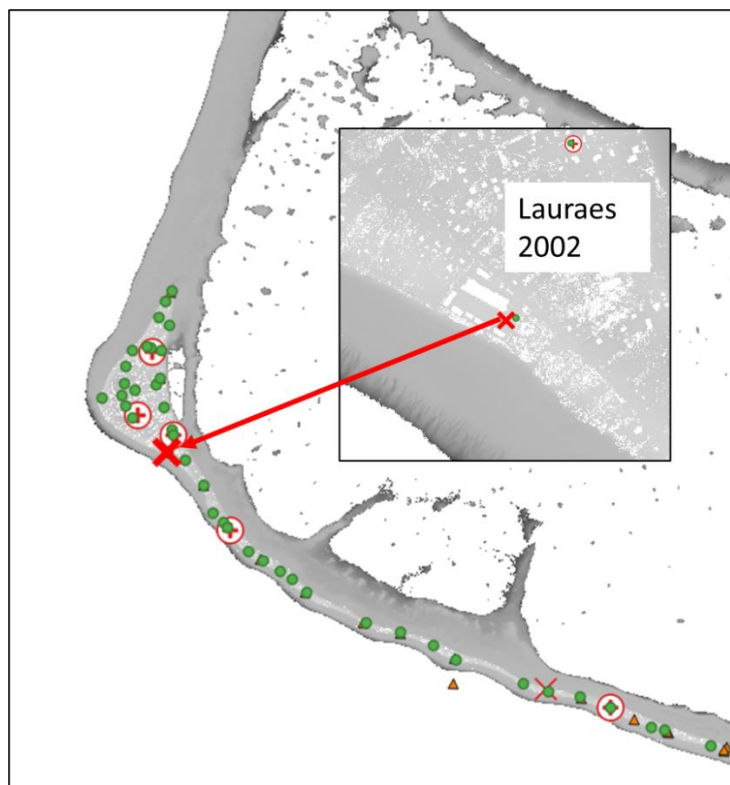


Figure 8: Lauraes 2002

| ID         | Survey | STN         | Ortho Ht (MGVD69) | Ell Ht (AusPos) | LiDAR Ell Ht | N (Ell-Ortho) |
|------------|--------|-------------|-------------------|-----------------|--------------|---------------|
| 20140005   | 2014   | Lauraes2002 | 3.123m            | 32.334m         | 32.202m      | 29.211m       |
| 201600045  | 2015   | Lauraes2002 | 3.123m            | 32.334m         | 32.202m      | 29.211m       |
| 2016000163 | 2016   | PNL319      | 2.98m             | 31.926m         | 32.209m      | 28.946m       |

Table 5: Lauraes 2002 height comparison

## All Control

Figure 9 shows the difference between ellipsoid heights from the 2019 LiDAR compared to the 2014 and 2015 control. Figure 10 shows the same comparison for the 2016 control (including BM Panel). A consistent -0.3m to 0.5m vertical range was used for both plots to provide a visual comparison of the two results. The 2014 and 2015 control compared with the LiDAR within a range of 0m to -0.2m (20cm), with a mean of -7.2cm and standard dev of 4.6cm. The 2016 comparisons showed a range of approximately -0.1m to 0.4m (40cm), with a mean of 14.0cm and standard deviation of 14.0cm.

This indicated that the LiDAR compared well with the 2014 and 2015 control, both in mean offset and variability (standard deviation). The 2016 control had a higher mean offset and much more variability, when compared to the LiDAR.

The mean offset could be attributed to the ellipsoid datum. Of greater concern is the variability. Some of the variability can be attributed to the vertical location of the control, as compared to the LiDAR spot. However, similar variability would be expected in all datasets and not greater for one than the other. The 2016 control variability is 3 times greater than the 2014 and 2015 control variability. This result puts more confidence in the 2014 and 2015 ellipsoid heights, over the 2016 ellipsoid heights.

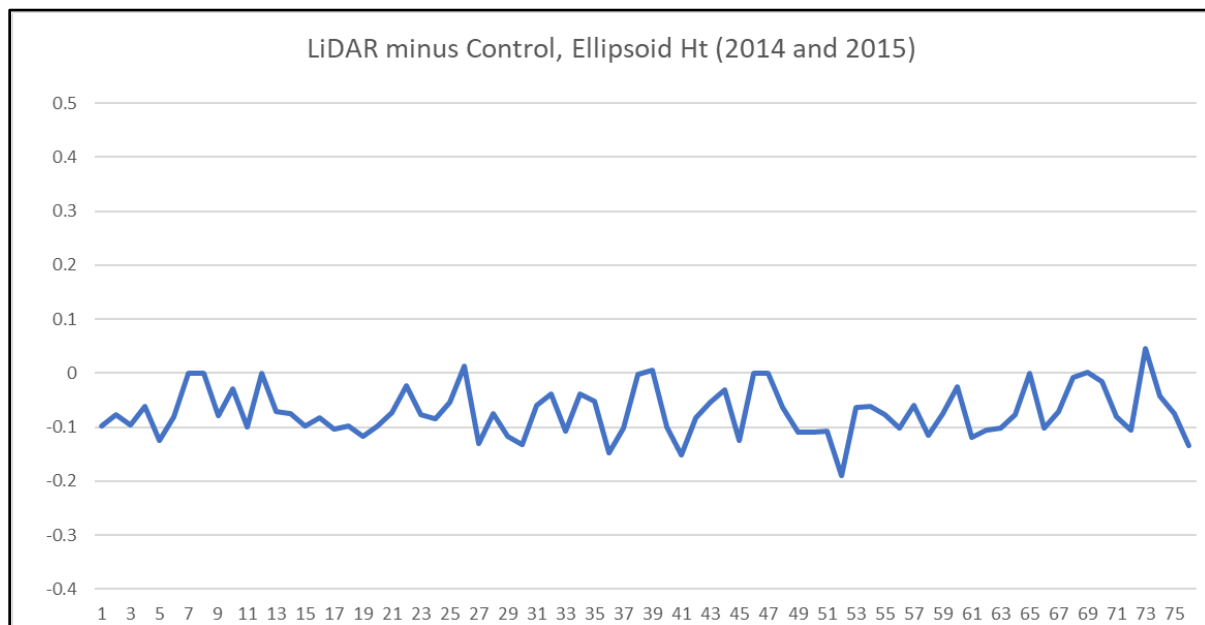


Figure 9: LiDAR minus control, ellipsoid heights, 2014 and 2015 control

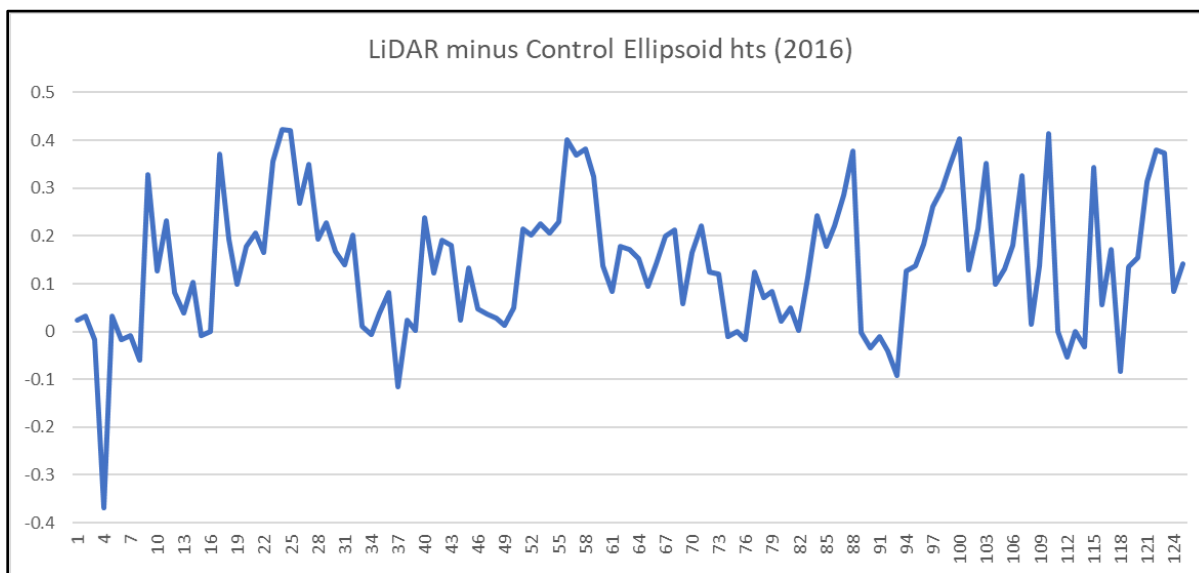


Figure 10: LiDAR minus control, ellipsoid heights, 2016 control

## 4 Majuro Geoid

A geoid is used to represent the surface of the earth based on a surface of constant gravity. A geoid model is used to represent this surface, relative to a reference ellipsoid. In order to create a local geoid, two datums must be defined; the ellipsoid reference and the orthometric reference, where we define the surface of constant gravity. A geoid model changes spatially because of gravity variations due to local variations in the earth's mass (e.g. mountains, seamounts, density of crust material). The model also varies because of the shape of the ellipsoid. A reference ellipsoid is a mathematical representation of the earth's surface and does not vary due to gravity.

Figure 11 depicts the relationship between the ellipsoid and geoid. This relationship is defined through the following equation:

$$h = N + H$$

Where:

- $h$  = ellipsoid height
- $H$  = orthometric height
- $N$  = geoid height (or geoid undulation)

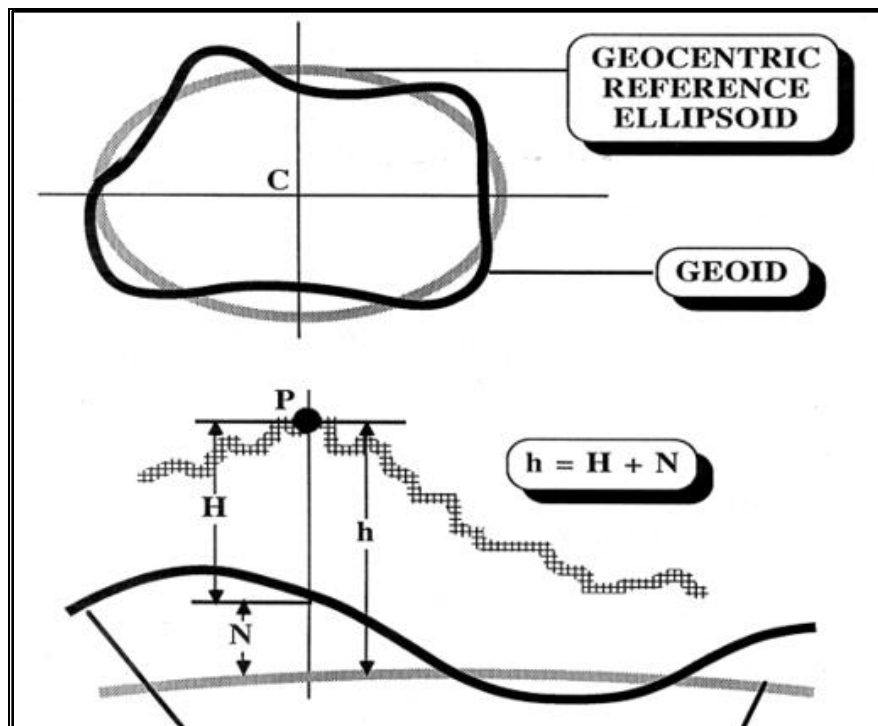


Figure 11: Geoid and ellipsoid relationship

One version of a geoid model (model of N values) is the Earth Gravity Model of 2008 (EGM08), created and distributed by the US National Geospatial-Intelligence Agency (NGA), see:

[https://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/egm08\\_wgs84.html](https://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/egm08_wgs84.html).

This model was developed to best represent the geoid, globally, relative to the WGS84 reference frame (and ellipsoid).

The original Majuro geoid was developed from the 2016 survey. Due to some uncertainty in the ellipsoid heights of control points, it was decided to abandon this geoid and develop a new one based on EGM08.

For the development of a local geoid model for Majuro (Mgeoid\_ITRF14ep2016\_MGVD17), the EGM08 model was vertically shifted to match the geoid height (N) at the tide gauge holding BM MAR13. The orthometric height was relative to MGVD17 and the ellipsoid height was relative to ITRF2014 epoch 2016, as established through AusPos at MAJU. The resulting geoid was validated by comparing the observed geoid ( $N=h-H$ ) to the geoid model.

Figure 12 shows the vertical relationship between the reference ellipsoid, MGeoid2020, EGM08, MGVD17, TGZ and several primary BMs and the MAJU CORS site.

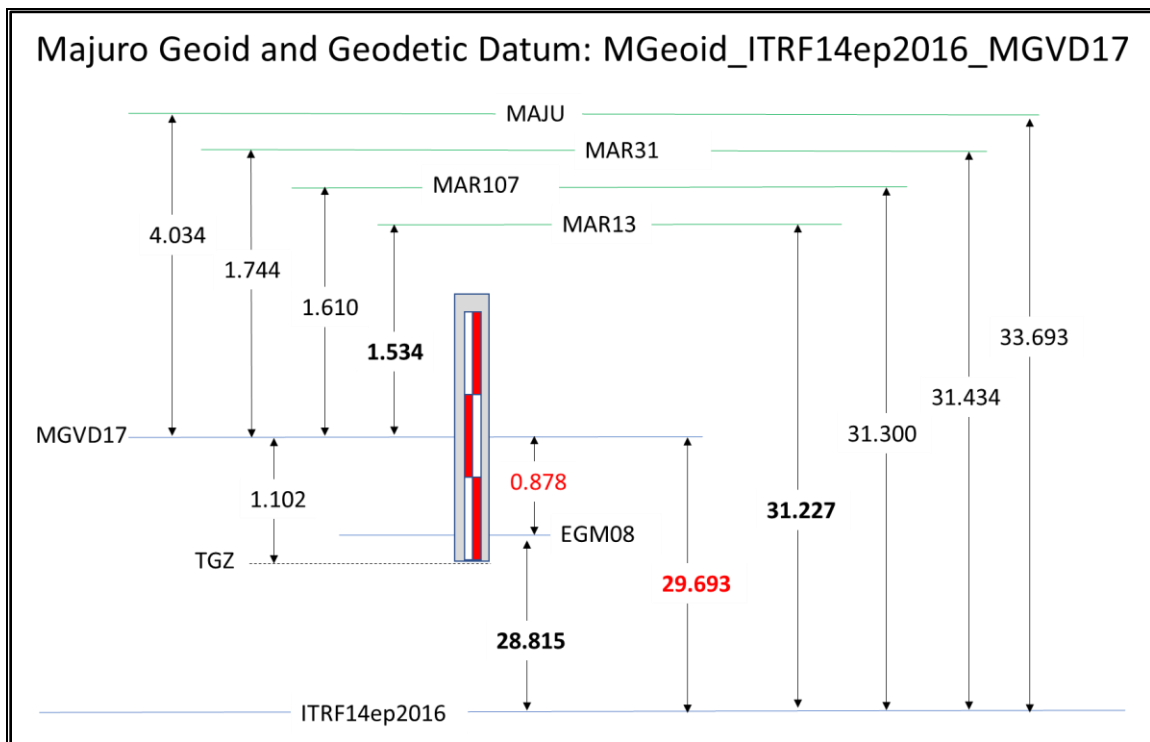


Figure 12: Majuro Geoid and Geodetic Datum, MGeoid\_ITRF14ep2016\_MGVD17

Figure 13 shows the difference between the modelled geoid values and the observed geoid values at the 2014 and 2015 control points. The mean offset was -2.6cm with a standard deviation of 4.6cm.

Figure 14 shows difference between the modelled geoid values and the observed geoid values at the 2016 control points. The mean offset was 16.7cm with a standard deviation of 16.3cm. A consistent -0.3m to 0.5m vertical range was used for both plots to provide a visual comparison of the two results.

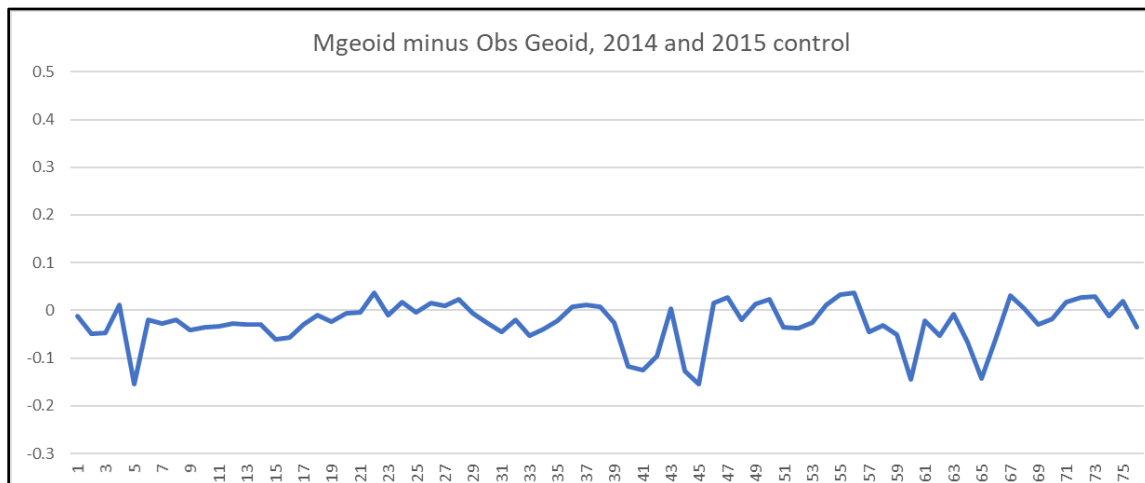


Figure 13: MGeoid2020 minus observed geoid values at 2014 and 2015 control points.

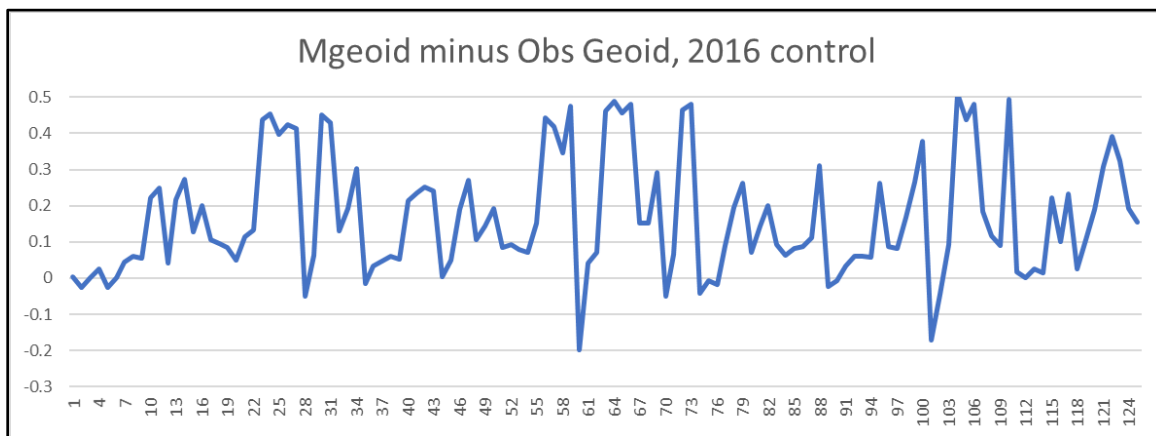


Figure 14: MGeoid2020 minus observed geoid values at 2016 control points.

These results are very similar to the ellipsoid height comparisons shown in Figure 9 and Figure 10. This indicates that the orthometric heights in all surveys are internally consistent, and as a result, the issue is likely with the ellipsoid heights of the 2016 control.

As discussed earlier, the change in the geoid model should be very gradual, and the variability between observed and modeled geoid should be very small, as long as all height measurements (orthometric and ellipsoid) are error free. The high variability in the observed geoid values from the 2016 control would indicate an issue with the 2016 measurements, and most likely the ellipsoid heights.

## 5 Conclusions and Recommendations

As part of its conduct of Quality Assurance on RMI LiDAR Data, IIC investigated the source of an approximately 1m vertical offset between control from four surveys between 2014 and 2017. The resulting report (*IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020*) showed the primary cause of the discrepancy was the 2016 survey's adopted coordinate for the continuously operating GNSS reference station MAJU. It also identified that the geoid developed for Majuro was based on this 2016 survey and that additional inconsistencies existed.

Therefore, SPC requested IIC to further review both the documentation and data from the 2014 to 2017 surveys in order to identify the issues and establish a revised geoid for Majuro; the findings of which are contained in this Report.

Based on assessment of the available data, analysis contained within this report and the subsequent validation, it is determined that the MGeoid2020 geoid model created from an adjusted EGM08 model provides a local geoid consistent with the 2014, 2015 and 2017 surveys. **It is recommended that this geoid be utilised for future survey and geodetic work requiring geoidal values.**

It is further recommended that validation surveys be conducted at critical control points in the southern and western areas when possible in order to physically confirm this recommended geoid. Also, more information on the origin of orthometric heights in the southern and western areas should be gathered.

It is noted that this Report builds on the findings and recommendations in the *IIC Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands dated 11 May 2020*. Therefore, it is recommended that both documents be read in conjunction.

## 6 References

- IIC Technologies (2020). Independent Review: Height Discrepancy at Majuro Atoll, Republic of Marshall Islands. Unpublished report written by IIC Technologies for the Pacific Community (SPC), May 2020.
- Fugro Report of Survey (2019). Acquisition and Processing of Lidar Data and Imagery for the Republic of the Marshall Islands. Report written by Fugro Australia Marine for the Republic of the Marshall Islands, October 2019.
- SPSLCMP (2009). South Pacific Sea Level and Climate Monitoring Project (SPSLCMP), Survey Report, EDM Height Traversing, Levelling Survey, Majuro, Marshall Islands, April 2009.
- SONEL (2019). MAJU GNSS CORS Station information website.  
<https://www.sonel.org/spip.php?page=gps&idStation=2055>